Detecting AI-Generated Text Using Lightweight Linguistic Models

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Introduction

As LLMs like ChatGPT grow more capable, distinguishing AI-generated text from human writing becomes increasingly difficult — yet increasingly important for academic integrity, journalism, and trust online.

We ask: **can a simple, interpretable model do the job?** Rather than using a heavyweight neural detector, we train a logistic regression on TF-IDF features — a model whose decisions can be fully audited and explained.

Research Questions:

- Can lightweight models reliably detect AI-generated text?
- Which linguistic patterns most distinguish ChatGPT from humans?
- Does detection hold across different topic domains?

Accuracy by Source

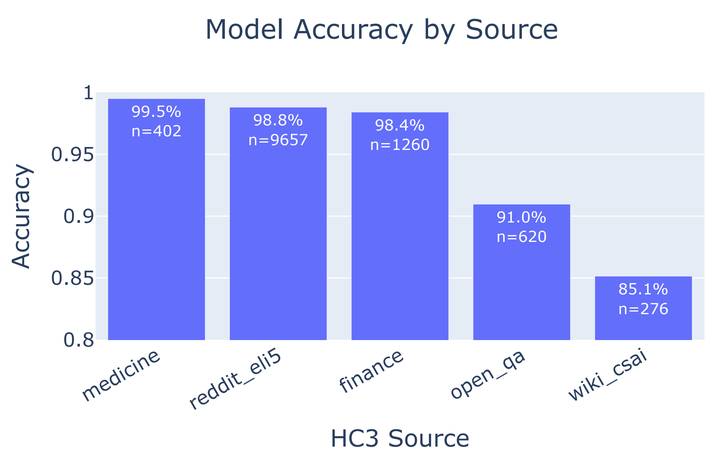

Figure 1. All six HC3 domains exceed 93% accuracy. Consistent performance across Reddit, StackExchange, medicine, and finance confirms the model learned **authorship style** — not domain-specific vocabulary.

Methodology

Dataset — HC3:

- Human and ChatGPT answers to the same questions
- Domains: Reddit, StackExchange, medicine, finance
- Split **by question group** to prevent leakage

Model Pipeline:

- TF-IDF (50k features, unigrams + bigrams)
- Logistic Regression with balanced class weights
- Fully interpretable coefficients

Model Confidence Distribution

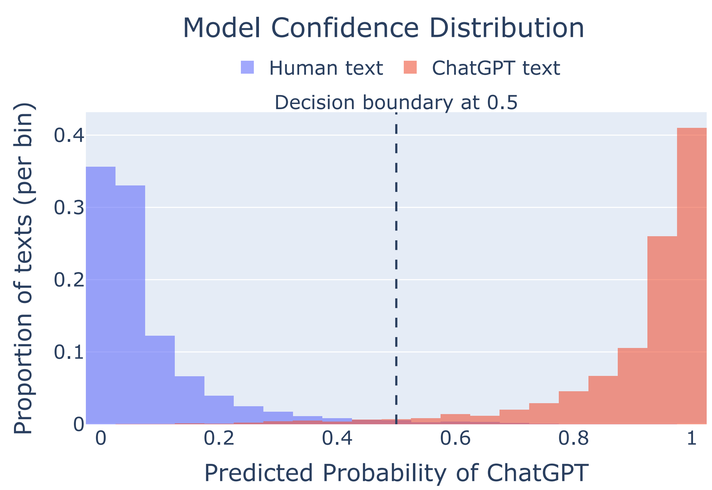

Figure 2. 97.4% ChatGPT correctly classified. 1.6% human misclassified Human texts. (blue) pile up near 0; ChatGPT texts (orange) pile up near 1. The sharp separation at the 0.5 boundary shows the model assigns **high-confidence predictions** to nearly every sample — reflecting ChatGPT's stylistic uniformity versus human variability.

Feature Importance

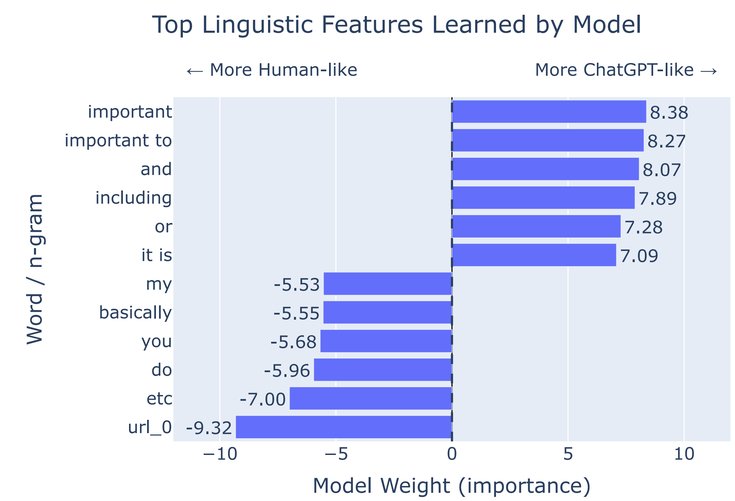

Figure 3. ChatGPT-indicative features (right) are formulaic phrases: *"in conclusion," "it is important to," "as an AI."* Human-indicative features (left) reflect informal, varied language. These poles explain why a linear model succeeds.

Findings

- >**95% accuracy** and ROC-AUC > 0.98 on held-out test data
- ChatGPT text is **stylistically uniform**; human text is diverse
- A handful of formulaic phrases are near-**exclusive signals** of LLM output
- Performance generalizes strongly across all six source domains

Takeaways:

- Lightweight models **match neural detectors** at a fraction of the cost
- Interpretability lets us **explain every decision** — a key advantage for real-world use
- As LLMs improve, lexical patterns will weaken; future detectors will need stylometric and syntactic features

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